DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot C

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-C01: Bus ridership [20 marks]

Scenario: You are the analyst for a city bus depot. The series riders (Riders per day) is recorded daily. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 7 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(6053)
m <- 7                    # observations per cycle (daily data)
n <- m * 10               # 10 full cycles
t <- 1:n
trend <- 5395.4 + 105.098 * t
riders <- round(trend * (1 + 0.22 * cos(2 * pi * (t - 2) / m)) + rnorm(n, 0, 224.6), 0)
x <- ts(riders, frequency = m, start = c(1, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(riders ~ 1:n) and report the slope per day). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 7 seasonal values). Name the peak day and the trough day with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 10 cycles is the highest value in the peak day from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(riders ~ 1:n))) and compute acf(d, lag.max = 7, plot = FALSE) using observation lags. Report the autocorrelation at lag 7 and at lag 3, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of riders in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per day [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

Code of Conduct: I certify that this work is my own. Signature: _____

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A) - Lokesh Kumar
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1.

A)

- Cycle length = 7 observations (days).
- No of full cycles = 10.
- Total observations = 7×10 = 70.
- The overall trend is rising because the trend equation has a positive slope of 105.098 riders per day.

B)

A)

The seasonal pattern repeats every 7 days. The seasonal component is extracted using decompose(x) and a figure.

The theotical seasonal pattern reaches its peak around day 2 of the cycle and a trough around 5-6 days.

Seasonal Swing = Peak Value - Trough Value

- Peak day : Day 2 , Value = 2178.0390
- Trough day : Day 6, Value = -1860.1356
- Seasonal Swing : 4038.1746

3)

A)

A seasonal plot is used to compare the 10 cycles. The pattern is considered consistent if the peak and trough to occur at approximately the same positions in each cycle.

```
cycle <- rep(1:10, each=7)
day <- rep(1:7, times=10)
Peak_day <- sapply(1:10, function(i){
    which.max(riders[cycle==i])
})
sum(peak_day==2)
```

4.)

A) Detrending removes the long-term increase before calculating auto correlation.

```
d <- resid(lm(riders~t))
a <- acf(d, lag.max=7, plot=FALSE)
a$acf[8]
a$acf[4]
2/sqrt(n)
```

The significance bound is approximately ±0.239.

- Lag 7 measures the correlation between observations one full cycle apart.
- Lag 3 measures the correlation btw observations 3 days apart.
- A lag-7 auto correlation greater than the significance bound in absolute value supports the presence of weekly seasonality.

5.

```
first_range <- diff(range(riders[1:7]))
last_range <- diff(range(riders[64:70]))

first_range
last_range
last_range/first_range
fit <- decompose(X, type = "multidaver")
adjustive <- riders/fit$figure[cycle]
coff(lm(adjusted ~t))[1]
```